Advanced Mathematical Techniques (AMT) Notes

1 Today's stuff

All practice today. Let's do this integral:

1.1 Problem 1

$$\int_0^1 \frac{\ln^2(1-t)}{t^2} dt$$

1.1.1 Check integral convergence

This integral converges. The logarithmic term has an expansion that starts at t, which means that the square of it starts at t^2, which cancels with the denominator, so all good. At $t = 1$, it's a logarithmic irregularity, so we can call it a weakling and move on.

1.1.2 Consider instead...

...the integral:

$$\int_0^1 t^{-2+\delta}(1-t)^{\varepsilon} dt$$

The δ, ε are there so we can put up a wall to prevent the divergence from destroying everything. Taking $\delta = 0$ would give us a divergence which makes everything fall apart, since allowing one thing to diverge will cause divergence to spread like a fire and burn the field.

We need to introduce a regulator to prevent the divergence from burning up the field long enough for us to extract the information about well defined integrals.

Now we evaluate the integral by the beta function definition:

$$\int_0^1 t^{-2+\delta}(1-t)^{\varepsilon} dt = \frac{\Gamma(-1+\delta)\Gamma(1+\varepsilon)}{\Gamma(\delta+\varepsilon)}$$

Now, this integral is enshrined in the coefficient of ε^2 of the considered integral. However, we need to make it $\Gamma(1+stuff)$ to have an expansion that doesn't break, so we'll use the **recurrence relation** to make the $\Gamma(1+stuff)$. Further, even though in our new integral we need $\delta > 1$, we don't really care because we found in our initial integral that it converges, so we can lowk ignore that pat of the new integral.

$$\frac{(\varepsilon+\delta)\delta(-1+\delta)\Gamma(-1+\delta)\Gamma(1+\varepsilon)}{\delta(-1+\delta)(\varepsilon+\delta)\Gamma(\varepsilon+\delta)} = -\frac{(\varepsilon+\delta)}{\delta(1-\delta)}$$

Thus,

$$\int_0^1 t^{-2+\delta}(1-t)^{\varepsilon} dt = -\frac{\epsilon+\delta}{\delta} \cdot (1+\delta+\delta^2+\delta^3+\cdot) \exp\left(\sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} \left[\delta^k + \varepsilon^k - (\varepsilon+\delta)^k\right]\right)$$

$$= \frac{\varepsilon+\delta}{\delta}(1+\delta+\delta^2+\cdots) \exp\left(\sum_{k=2}^{\infty} \cdot - 2\varepsilon\delta\right) = \frac{\varepsilon+\delta}{\delta}(1+\delta+\delta^2+\cdots) \exp\bigl(-\zeta(2)\varepsilon\delta + \zeta(3)(\varepsilon^2\delta+\varepsilon\delta^2)+\cdots\bigr)$$

$$= -\frac{1}{\delta}(\varepsilon+\delta)(1+\delta+\delta^2+\delta^3+\cdots) \exp\bigl(-\zeta(2)\varepsilon\delta + \zeta(3)(\varepsilon^2\delta+\varepsilon\delta^2)+\cdots\bigr)$$

Considering the epsilon from the first term, we need an epsilon delta from the rest. Considering the delta, we can't get a ε^2 without a delta so we hate it for not doing anything worthwhile. Breaking down the exponential, we're only going to get the right term from the epsilon-delta term in the beginning, so we only care about the first power. This makes what we need just $\zeta(2)$.

Doing the same for the logarithm cubed,